Project 1
Final Presentation

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BIOS 6624

Research Questions

- Q1 - How does treatment response differ between participants who reported hard drug use at baseline and those who reported no hard drug use at baseline?
  - Measurements of Treatment Response: Viral Load, CD4+ T Cell Count, Mental QOL Score, Physical QOL Score
- Q2 - Does adherence account for some of the difference between these two groups of participants?

Characteristic	Overall N = 446[1]	No Hard Drug Use N = 410[1]	Hard Drug Use N = 36[1]
Viral Load (copies/mL)	1,089,914 (11,946,680)	1,168,144 (12,457,634)	198,958 (445,061)
CD4+ T Cell Count (# of cells)	373 (203)	374 (204)	360 (201)
Physical QOL Score	51 (9)	52 (9)	49 (7)
Mental Health QOL Score	45 (13)	45 (14)	42 (12)
Age (years)	43 (9)	43 (9)	44 (10)
BMI (kg/m^2)	25.2 (4.4)	25.3 (4.4)	23.6 (3.4)
Race			
Other	161 (36%)	144 (35%)	17 (47%)
White, Non-Hispanic	285 (64%)	266 (65%)	19 (53%)
Highest Education Reached			
Not College Graduate	252 (57%)	226 (55%)	26 (72%)
College Graduate	194 (43%)	184 (45%)	10 (28%)
Smoking Status			
Not Current Smoker	275 (62%)	266 (65%)	9 (25%)
Current Smoker	171 (38%)	144 (35%)	27 (75%)
Income			
Less than $10,000	95 (21%)	81 (20%)	14 (39%)
$10,000–19,999	44 (9.9%)	41 (10%)	3 (8.3%)
$20,000–29,999	79 (18%)	76 (19%)	3 (8.3%)
$30,000–39,999	64 (14%)	57 (14%)	7 (19%)
$40,000–49,999	48 (11%)	48 (12%)	0 (0%)
$50,000–59,999	80 (18%)	71 (17%)	9 (25%)
$60,000 or more	36 (8.1%)	36 (8.8%)	0 (0%)

[1] Continuous variables: Mean (SD); Categorical variables: n (%)

Data Overview

- Initial 3,935 observations from 715 unique participants
- Exclusions:
  - Missing baseline or year 2 visit
  - Biologically Implausible BMI
  - Missing reported income
  - Missing any outcomes of interest

Analysis

Identical Model Structures

- Outcomes - Viral Load, CD4+ T Cell Count, Mental QOL Score, Physical QOL Score at Year 2
- Primary Predictor - Hard Drug Use at Baseline
- Covariates - Outcomes at Baseline, Age, BMI, Race, Education, Smoking Status, Adherence

Frequentist Approach

- Linear Regression Model

Bayesian Approach

- CmdStanR
- Weakly Informative Priors

  Coefficients ~ Normal(0, 100)

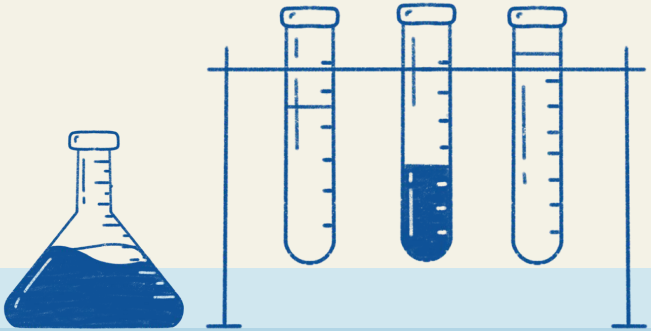
  Sigma ~ Half-Normal(0, 100)
- 4 chains, 25,000 iterations, 5,000 warm-up

Results

	Frequentist			Bayesian		
Outcome	β	95% CI	p-value	Posterior Mean	95% HPDI	P(β < 0)
Without Adherence						
Viral Load	-0.20	(-0.59, 0.2)	0.325	-0.20	(-0.59, 0.19)	0.837
T-Cell Count	-159.02	(-221, -97.04)	<0.001	-143.47	(-203, -85.22)	1.000
Mental QoL	0.06	(-3.5, 3.62)	0.974	0.05	(-3.44, 3.64)	0.489
Physical QoL	-3.28	(-6.01, -0.55)	0.019	-3.28	(-6.04, -0.57)	0.991
With Adherence						
Viral Load	-0.15	(-0.55, 0.24)	0.446	-0.15	(-0.54, 0.24)	0.775
T-Cell Count	-166.00	(-228.02, -103.98)	<0.001	-148.64	(-207.48, -89.83)	1.000
Mental QoL	-0.22	(-3.79, 3.36)	0.905	-0.22	(-3.81, 3.41)	0.549
Physical QoL	-3.47	(-6.21, -0.72)	0.013	-3.46	(-6.23, -0.72)	0.993

Viral load was log10 transformed prior to analysis.

Discussion

- Baseline hard drug use associated with clinically and statistically significant reductions in CD4+ T cell count (~159 cells, threshold: 50) and physical QoL (~3.28 points, threshold: 2)
- No significant or clinically meaningful associations observed for viral load or mental QoL
- Adjusting for adherence did not attenuate the hard drug use effect — coefficients remained largely unchanged
- Near-identical estimates across both approaches — results robust to choice of analytical framework

Questions?